Centre Number	Candidate Number	Candidate Name

NAMIBIA SENIOR SECONDARY CERTIFICATE

CHEMISTRY ORDINARY LEVEL 6117/2

PAPER 2 Structured Questions 1 hour 30 minutes

Marks 80 **2024**

Additional Materials: Non-programmable calculator
Ruler

INSTRUCTIONS AND INFORMATION TO CANDIDATES

- Candidates answer on the Question Paper in the spaces provided.
- Write your Centre Number, Candidate Number and Name in the spaces provided at the top of this page.
- Write in dark blue or black pen.
- You may use a soft pencil for any diagrams, graphs or rough working.
- Do not use correction fluid.
- Do not write in the margin *For Examiner's Use*.
- Answer **all** questions.
- The number of marks is given in brackets [ ] at the end of each question or part question.
- You will lose marks if you do not show your working or if you do not use appropriate units.
- The Periodic Table is printed on page 15.

For Examiner's Use	
1	
2	
3	
4	
5	
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7	
Total	

Marker	
Checker	

This document consists of **15** printed pages and **1** blank page.

Republic of Namibia

MINISTRY OF EDUCATION, ARTS AND CULTURE

For Examiner's Use

1 Fig. 1.1 shows part of the Periodic Table. Only some of the elements are shown.

The letters in the table do not represent the symbols of the elements.

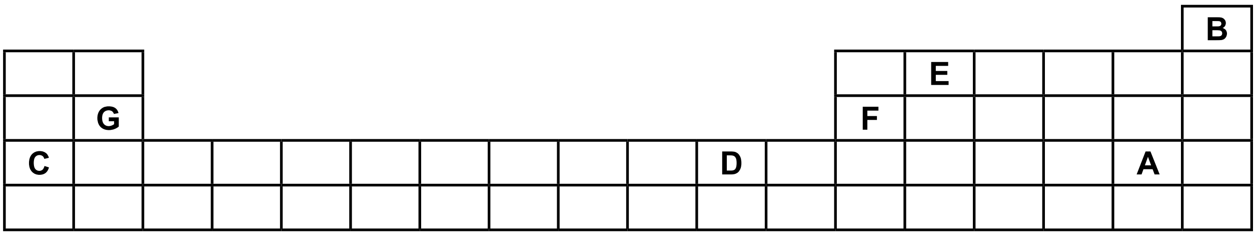


Fig 1.1

(a) Use the Periodic Table to answer the questions.

Each element may be used once, more than once or not at all.

Choose, from **A** to **F**, the element which

(i) is in the same period as calcium,

.... [1]

(ii) is a liquid at room temperature which contains diatomic molecules,

.... [1]

(iii) forms an amphoteric oxide,

.... [1]

(iv) forms a soluble hydroxide with water,

.... [1]

(v) is used to fill weather balloons.

.... [1]

(b) **D** is a transition element. Transition elements are harder and stronger than Group I elements.

Describe **two** other properties of transition elements which are different from those of Group I elements.

1 ....

2 .... [2]

[7]

For Examiner's Use

2 Magnesium forms the ion, Mg^{2+} and bromine forms the ion, Br^-.

These two ions form magnesium bromide.

(a) State the number of electrons and neutrons in a bromide ion.

number of electrons ....

number of neutrons .... [2]

(b) Describe how a magnesium ion, Mg^{2+}, is formed from a magnesium atom, Mg.

.... [1]

(c) Magnesium bromide is an ionic compound.

Draw dot-and-cross diagrams to show the structure of the ions in magnesium bromide.

Show outer shells only.

[3]

(d) Electricity can be used to break down aqueous or molten ionic compounds such as magnesium bromide.

(i) Name the process which uses electricity to break down aqueous or molten ionic compounds.

.... [1]

(ii) Explain why the ionic compound needs to be aqueous or in a molten state.

....

.... [1]

(iii) In the electrolytic cell, at the cathode, magnesium ions, Mg^{2+}, are changed into magnesium atoms. At the anode, bromide ions, Br^-, are changed into bromine molecules.

Write the ionic half equations for the reactions at the cathode and at the anode.

cathode ....

anode .... [2]

For Examiner's Use

(e) Both magnesium bromide and diamond form lattices.

(i) Distinguish between the lattice of magnesium bromide and that of diamond.

....

....

....

.... [2]

(ii) Describe the structure of diamond.

....

.... [1]

(iii) In terms of bonding, explain why diamond has a high melting point.

....

.... [1]

[14]

For Examiner's Use

3 Ammonia is manufactured by the Haber process. When nitrogen is reacted with hydrogen in a closed container, an equilibrium mixture is formed. An iron catalyst is used and the reaction is exothermic.

The chemical equation is shown.

$$N_2 + H_2 \rightleftharpoons NH_3$$

(a) Balance the chemical equation.

N_2 + ...H_2 $\rightleftharpoons$ .....NH_3 [1]

(b) State why a catalyst is used.

.... [1]

(c) Explain, in terms of bond breaking and bond forming, why the reaction is exothermic.

....

....

.... [2]

(d) Describe and explain the effect on the position of equilibrium when the pressure is decreased.

....

....

....

.... [2]

(e) State the shape of a molecule of ammonia.

.... [1]

(f) Ammonia is a base.

Name the ion present in the solution to make it alkaline.

.... [1]

For Examiner's Use

(g) Ammonium phosphate, a fertiliser can be formed from ammonia (NH_3) and phosphoric acid (H_3PO_4) as shown in the equation.

$$3NH_3 + H_3PO_4 \rightarrow (NH_4)_3PO_4$$

2.891 g of phosphoric acid, H_3PO_4 reacts with ammonia, NH_3.

(i) Calculate the number of moles in 2.891 g of phosphoric acid, H_3PO_4.

Show your working.

number of moles =................ mol [2]

(ii) Calculate the mass of ammonia, NH_3 that reacted with 2.891 g of phosphoric acid, H_3PO_4.

Show your working.

mass =.................. g [2]

(iii) State **two** other elements present in fertilisers other than phosphorus.

1 ....

2 .... [2]

[14]

For Examiner's Use

4 Fig. 4.1 shows two test-tubes each containing dilute hydrochloric acid. A piece of Universal Indicator paper is added to test-tube **X** and magnesium ribbon to test-tube **Y**.

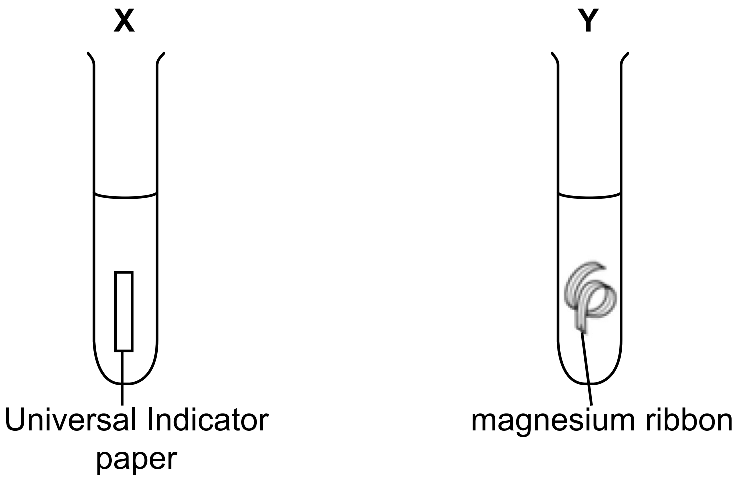


Fig. 4.1

(a) Describe what would be observed in each test-tube.

test-tube **X** ....

.... [1]

test-tube **Y**..

.... [1]

(b) Hydrochloric acid is a strong acid.

Explain the difference between strong and weak acids in terms of the dissociation into ions.

....

.... [1]

(c) Acids react with alkalis in a process called neutralisation to give salt and water.

(i) Describe the chemical test for water.

test ....

result ....

.... [2]

(ii) Give the name of the salt formed when calcium hydroxide neutralises sulfuric acid.

.... [1]

For Examiner's Use

(d) Silver chloride, ($AgCl$), an insoluble salt, can be prepared in the laboratory.

(i) State the name of the method used to prepare insoluble salts.

.... [1]

(ii) Suggest **two** suitable starting materials for the preparation of silver chloride.

1 ....

2 .... [2]

[9]

For Examiner's Use

5 A list of elements is shown.

zinc	**copper**	**hydrogen**	**carbon**	**barium**
calcium	**magnesium**	**iron**	**aluminium**	**sodium**

(a) Use the list of elements to answer the questions.

Each element may be used once, more than once or not at all.

Identify

(i) a metal which burns with a bright yellow flame,

.... [1]

(ii) the least reactive metal,

.... [1]

(iii) the non-metal component of steel,

.... [1]

(iv) a metal which is extracted by reduction of its ore using carbon,

.... [1]

(v) a metal which is used to galvanise iron.

.... [1]

(b) Compare **one** physical property of metals and non-metals.

....

.... [1]

(c) Aluminium is a widely used metal.

(i) Explain why in an experiment, a sample of aluminium appeared less reactive than expected.

....

.... [1]

(ii) Name the main ore of aluminium.

.... [1]

For Examiner's Use

(d) When zinc granules are added to aqueous copper(II) sulfate, a reaction occurs. During the reaction, a red-pink solid is formed and the solution becomes colourless.

(i) Name the red-pink solid and the colourless solution.

red-pink solid ....

colourless solution .... [2]

(ii) Explain why a red-pink solid and a colourless solution are formed.

....

....

....

.... [2]

(iii) In terms of changes in ionic charge, state which substance is oxidised.

.... [1]

[13]

For Examiner's Use

6 Ethanol is an alcohol. Alcohols are a 'family' of organic compounds which have the same general formula.

(a) What is the name given to any 'family' of organic compounds which have the same general formula and similar chemical properties?

.... [1]

(b) Give the general formula of alcohols.

.... [1]

(c) Ethanol can also be manufactured by the fermentation of glucose, $C_6H_{12}O_6$.

(i) State **two** conditions needed for the fermentation of glucose.

1 ....

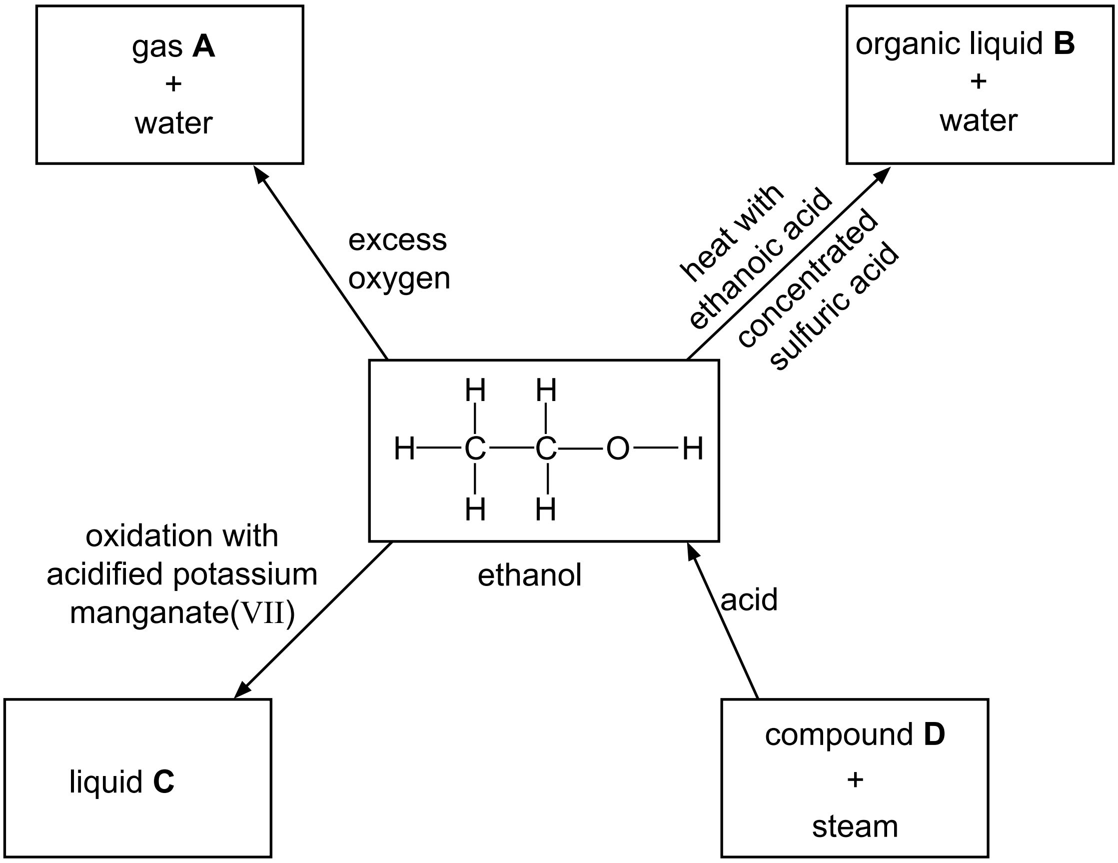
2 .... [2]

(ii) Complete and balance the chemical equation for the fermentation of glucose.

$C_6H_{12}O_6 \rightarrow$ ...... C_2H_5OH + ... [2]

For Examiner's Use

(d) The diagrams show some reactions of ethanol.



(i) Give the names of **A**, **B** and **C**.

A..

B..

C.. [3]

(ii) Deduce the identity of compound **D**, which reacts with steam to form ethanol.

.... [1]

For Examiner's Use

(e) A simple sugar unit can be represented as shown.

Simple sugars can be polymerised to make more complex carbohydrates.

(i) Draw a diagram to show part of a carbohydrate polymer made from the simple sugar shown.

[2]

(ii) Name the chemical process which occurs when a carbohydrate polymer is broken down into simple sugars.

.... [1]

(iii) What conditions are needed for the process in **part (e)(ii)** to occur?

.... [1]

[14]

For Examiner's Use

7 The air around us is a mixture of gases.

(a) Nitrogen, oxygen and other substances are found in clean, dry air.

(i) Other than nitrogen and oxygen, give another element found in clean, dry air.

.... [1]

(ii) Nitrogen and oxygen can be separated from liquid air.

State the name of this process.

.... [1]

(b) Give **one** use of oxygen.

.... [1]

(c) Some of the pollutants in air are carbon monoxide and lead compounds.

(i) Give **one** effect of each of these pollutants on health.

carbon monoxide ....

....

lead compounds..

.... [2]

(ii) Sulfur dioxide is another pollutant of the air which can be used for the manufacture of sulfuric acid.

State the conditions used in this industrial process. Include units.

....

.... [2]

(iii) Name **one** other pollutant present in air. State the source of this pollutant.

pollutant ....

source of pollutant.. [2]

[9]

DATA SHEET
The Periodic Table of the Elements

Group																	
I	**II**											**III**	**IV**	**V**	**VI**	**VII**	**0**
							1 **H** Hydrogen 1										4 **He** Helium 2
7 **Li** Lithium 3	9 **Be** Beryllium 4											11 **B** Boron 5	12 **C** Carbon 6	14 N Nitrogen 7	16 **O** Oxygen 8	19 **F** Fluorine 9	20 **Ne** Neon 10
23 **Na** Sodium 11	24 **Mg** Magnesium 12											27 **A*l*** Aluminium 13	28 **Si** Silicon 14	31 **P** Phosphorus 15	32 **S** Sulfur 16	35,5 **C*l*** Chlorine 17	40 **Ar** Argon 18
39 **K** Potassium 19	40 **Ca** Calcium 20	45 **Sc** Scandium 21	48 **Ti** Titanium 22	51 **V** Vanadium 23	52 **Cr** Chromium 24	55 **Mn** Manganese 25	56 **Fe** Iron 26	59 **Co** Cobalt 27	59 **Ni** Nickel 28	64 **Cu** Copper 29	65 **Zn** Zinc 30	70 **Ga** Gallium 31	73 **Ge** Germanium 32	75 **As** Arsenic 33	79 **Se** Selenium 34	80 **Br** Bromine 35	84 **Kr** Krypton 36
85 **Rb** Rubidium 37	88 **Sr** Strontium 38	89 **Y** Yttrium 39	91 **Zr** Zirconium 40	93 **Nb** Niobium 41	96 **Mo** Molybdenum 42	**Tc** Technetium 43	101 **Ru** Ruthenium 44	103 **Rh** Rhodium 45	106 **Pd** Palladium 46	108 **Ag** Silver 47	112 **Cd** Cadmium 48	115 **In** Indium 49	119 **Sn** Tin 50	122 **Sb** Antimony 51	128 **Te** Tellurium 52	127 ***I*** Iodine 53	131 **Xe** Xenon 54
133 **Cs** Caesium 55	137 **Ba** Barium 56	139 **La** Lanthanum 57 *	178 **Hf** Hafnium 72	181 **Ta** Tantalum 73	184 **W** Tungsten 74	186 **Re** Rhenium 75	190 **Os** Osmium 76	192 **Ir** Iridium 77	195 **Pt** Platinum 78	197 **Au** Gold 79	201 **Hg** Mercury 80	204 **T*l*** Thallium 81	207 **Pb** Lead 82	209 **Bi** Bismuth 83	**Po** Polonium 84	**At** Astatine 85	**Rn** Radon 86
Fr Francium 87	226 **Ra** Radium 88	227 **Ac** Actinium 89 †															
***58 - 71 Lanthanoid series** **†90 - 103 Actinoid series**				140 **Ce** Cerium 58	141 **Pr** Praseodymium 59	144 **Nd** Neodymium 60	**Pm** Promethium 61	150 **Sm** Samarium 62	152 **Eu** Europium 63	157 **Gd** Gadolinium 64	159 **Tb** Terbium 65	162 **Dy** Dysprosium 66	165 **Ho** Holmium 67	167 **Er** Erbium 68	169 **Tm** Thulium 69	173 **Yb** Ytterbium 70	175 **Lu** Lutetium 71
Key	a **X** b	a = relative atomic mass **X** = atomic symbol b = proton (atomic) number		232 **Th** Thorium 90	**Pa** Protactinium 91	238 **U** Uranium 92	**Np** Neptunium 93	**Pu** Plutonium 94	**Am** Americium 95	**Cm** Curium 96	**Bk** Berkelium 97	**Cf** Californium 98	**Es** Einsteinium 99	**Fm** Fermium 100	**Md** Mendelevium 101	**No** Nobelium 102	**Lr** Lawrencium 103

The volume of one mole of any gas is 24 dm^3 at room temperature and pressure (r.t.p.).

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